CPTG: Structural-Collapse Interpretation of M31-2014-DS1

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Abstract

M31-2014-DS1 is a failed-supernova candidate in M31 whose observed record combines mid-infrared brightening, optical disappearance, persistent red infrared emission, molecular and dusty remnant structure, weak inferred ejection energy, and no detected X-ray source. Conventional interpretation treats the event as the collapse of an approximately twelve-to-thirteen solar-mass progenitor into an apparent stellar-mass black hole near five solar masses. This article presents a structural-collapse interpretation of the same system within Curvature Polarization Transport Gravity (CPTG). In CPTG, compact-object formation is governed by a finite-core structural condition involving both physical core mass and curvature-gradient organization. The remnant mass inferred from exterior gravitational behavior may therefore exceed the physical baryonic core mass when curvature polarization amplifies the apparent field and curvature transport organizes the surrounding remnant geometry. Applied to M31-2014-DS1, the conventional five-solar-mass remnant is interpreted as an apparent gravitational mass, while the physical finite core may lie in the representative CPTG band of approximately 1.1–1.4 solar masses. The remaining apparent mass is attributed to curvature-polarization support and organized transport structure rather than additional matter. This mass split remains compatible with the observed low-energy ejecta, dusty mid-infrared remnant, optical disappearance, and X-ray silence, and it gives clear falsifiable expectations for continued infrared, optical, and X-ray monitoring.

1 Motivation

M31-2014-DS1 is a nearby and unusually well-observed candidate for quiet massive-star collapse. The discovery analysis reports a hydrogen-depleted supergiant in the Andromeda galaxy that brightened in the mid-infrared in 2014 and then faded dramatically, becoming undetectable in optical light while also fading in total luminosity [1]. Follow-up work with JWST and Chandra reports a persistent red mid-infrared remnant, molecular absorption, dust features, continued bolometric fading, and no detected X-ray source [2]. A separate analysis emphasizes that the persistent infrared source, asymmetric dust geometry, and lack of X-rays leave the fate of the object observationally unsettled, with a dusty survivor or merger-like interpretation remaining viable [3].

CPTG [6] approaches this object from a different physical starting point. The central question is not whether M31-2014-DS1 formed a conventional exposed black hole of roughly five solar masses. The CPTG question is whether the observed event is consistent with structural horizon closure: a transition in which a finite compact core crosses a curvature-gradient threshold while its external gravitational appearance is amplified by curvature polarization and organized by curvature transport.

In the CPTG compact-object framework, quiet collapse, lower-mass-gap objects, apparent compact-object masses, hidden remnants, and vanishing-star events arise from the same gradient-bounded mechanism [4]. The relevant compact state is not defined by mass alone. It is defined by the physical core mass together with the local curvature-gradient state:

$$(M_{\mathrm{core}}, \chi_{\mathrm{grad}})\,. \tag{1}$$

This paper tests whether the published mass, luminosity, gas, dust, and X-ray constraints for M31-2014-DS1 remain coherent when the conventional remnant mass is treated as an apparent gravitational mass rather than as the physical baryonic core mass.

2 CPTG Structural-Collapse Framework

CPTG is a nonlinear geometric theory in which baryonic matter sources curvature, curvature develops a nonlinear polarization response, and organized curvature can be redistributed through transport [5]. The compact-object reduction begins from the same structural logic: the field response depends on both local field strength and spatial gradient organization.

The reduced curvature invariant is

$$C = g_j g_j + \kappa^2 \left(\partial_i g_j\right)\left(\partial_i g_j\right). \tag{2}$$

The corresponding dimensionless curvature-response variable is

$$\chi = \frac{C}{a_\star^2}, \tag{3}$$

with gradient contribution

$$\chi_{\text{grad}} = \frac{\kappa^2 (\partial_i g_j)(\partial_i g_j)}{a_\star^2}. \tag{4}$$

The structural horizon condition is then written as

$$\chi_{\text{grad}} \geq \chi_{\text{crit}}. \tag{5}$$

This condition states that compact-object formation is not controlled by enclosed mass alone. A low physical core mass can enter a horizon-like compact state if the gradient organization reaches the critical structural threshold. Conversely, mass without the required gradient organization is not sufficient by itself.

The finite-core interior is represented by a regular potential expansion,

$$\Phi(r) = \Phi_0 + \alpha_{\text{core}} r^2 + O(r^4), \tag{6}$$

which gives

$$g(r) = -\nabla\Phi(r) \simeq -2\alpha_{\text{core}} r, \tag{7}$$

and therefore

$$\lim_{r\to 0} g(r) = 0. \tag{8}$$

The core is therefore finite rather than singular. The exterior gravitational appearance is not forced to equal the physical baryonic core mass. CPTG writes the apparent mass relation as

$$M_{\text{apparent}} = A_{\text{pol}} M_{\text{core}}, \tag{9}$$

where A_{pol} is the apparent-mass amplification supplied by exterior curvature polarization. Curvature transport does not add matter. It governs the directional organization of the surrounding remnant geometry, dust, envelope, and fallback environment.

3 Observed Inputs

The observational anchor values used in this article are the published quantities most relevant to the CPTG structural-collapse interpretation. The discovery analysis identifies the event as a failed-supernova candidate associated with the disappearance of a massive star in M31 [1]. The follow-up JWST and Chandra analysis reports a progenitor consistent with approximately 12–13 solar masses, a conventional remnant interpretation near five solar masses, an extremely red JWST source, dust and molecular structure, continued bolometric fading, and no X-ray detection [2]. The alternate fate analysis detects a persistent infrared source and emphasizes that the late-time source may be a dust-enshrouded survivor or merger product rather than an unambiguous accreting black hole [3].

Observed quantity	Adopted value	Role in CPTG interpretation
Object	M31-2014-DS1	Failed-supernova / vanishing-star candidate
Host system	M31	Nearby resolved stellar-collapse target
Progenitor mass anchor	12–13 $M_\odot$	Initial stellar-scale context
Conventional remnant mass	Approximately 5 $M_\odot$	Apparent gravitational remnant mass
Bolometric remnant luminosity	$\log(L/L_\odot) \approx 3.88$	Late remnant energy scale
Molecular gas mass	Approximately 0.1 $M_\odot$	Ejected/remnant gas scale
Gas speed	Approximately 100 $\mathrm{km\,s^{-1}}$	Kinetic-energy scale
Dust-shell radius	Approximately 40–200 au	Remnant geometry and transport context
X-ray limit	$L_X \lesssim 1.5 \times 10^{35}$ erg s^{-1}	Accretion/fallback constraint
Late-time source status	Persistent red infrared source	Remnant/survivor discriminator

Table 1: Observed inputs used for the CPTG structural-collapse interpretation of M31-2014-DS1.

The central observational fact is that the object satisfies the qualitative sequence expected of a structural-threshold target:

$$\begin{gathered}\text{mid-infrared brightening}\\ \downarrow\\ \text{optical disappearance}\\ \downarrow\\ \text{persistent red infrared remnant}\\ \downarrow\\ \text{weak or absent X-ray emission.}\end{gathered} \tag{10}$$

This sequence does not by itself prove CPTG compact-object physics. It identifies M31-2014-DS1 as a high-value target for testing whether apparent remnant mass, physical core mass, luminosity suppression, and remnant geometry separate in the way CPTG predicts.

4 Mass Split and Apparent-Mass Amplification

The conventional interpretation takes the remnant mass to be approximately

$$M_{\mathrm{remnant,std}} \simeq 5M_\odot. \tag{11}$$

In the CPTG interpretation, this value is treated as the apparent gravitational mass:

$$M_{\text{apparent}} \simeq 5M_{\odot}. \tag{12}$$

The physical finite core is allowed to occupy the representative compact-object band

$$1.1M_{\odot} \leq M_{\text{core}} \leq 1.4M_{\odot}. \tag{13}$$

The required polarization amplification is therefore

$$A_{\text{pol}} = \frac{M_{\text{apparent}}}{M_{\text{core}}}. \tag{14}$$

For the adopted range, this gives

$$3.571 \leq A_{\text{pol}} \leq 4.545. \tag{15}$$

The geometric apparent-mass equivalent is

$$M_{\text{geom}} = M_{\text{apparent}} - M_{\text{core}}. \tag{16}$$

Thus,

$$3.6M_{\odot} \lesssim M_{\text{geom}} \lesssim 3.9M_{\odot}. \tag{17}$$

The fraction of the apparent mass supplied by geometric response is

$$f_{\text{geom}} = \frac{M_{\text{geom}}}{M_{\text{apparent}}}. \tag{18}$$

For the representative band,

$$0.72 \lesssim f_{\text{geom}} \lesssim 0.78. \tag{19}$$

A central reference case is

$$M_{\text{core}} = 1.25M_{\odot}. \tag{20}$$

For this case,

$$A_{\rm pol} = \frac{5.00}{1.25} = 4.00, \tag{21}$$

$$M_{\rm geom} = 3.75 M_\odot, \tag{22}$$

$$f_{\rm geom} = 0.75. \tag{23}$$

The corresponding interpretation is simple: one quarter of the apparent mass is physical finite-core mass, while three quarters is the mass-equivalent exterior gravitational appearance supplied by curvature polarization and organized transport structure.

Interpretation	Physical core	Apparent mass	Geometric contribution
Standard failed supernova	$5.0M_\odot$	$5.0M_\odot$	None
CPTG lower core	$1.1M_\odot$	$5.0M_\odot$	$3.9M_\odot$ equivalent
CPTG reference core	$1.25M_\odot$	$5.0M_\odot$	$3.75M_\odot$ equivalent
CPTG upper core	$1.4M_\odot$	$5.0M_\odot$	$3.6M_\odot$ equivalent

Table 2: Standard and CPTG mass interpretations for M31-2014-DS1.

5 Ejection Energy Closure

The reported molecular gas scale is approximately

$$M_{\rm gas} \simeq 0.1 M_\odot, \tag{24}$$

with characteristic speed

$$v_{\rm gas} \simeq 100\,\mathrm{km\,s^{-1}}. \tag{25}$$

The kinetic energy scale is

$$E_{\rm kin} = \frac{1}{2} M_{\rm gas} v_{\rm gas}^2. \tag{26}$$

Using

$$M_\odot = 1.98847 \times 10^{33}\,\mathrm{g}, \tag{27}$$

and

$$100\,\mathrm{km\,s^{-1}} = 10^7\,\mathrm{cm\,s^{-1}}, \tag{28}$$

one obtains

$$E_{\mathrm{kin}} = 9.94235 \times 10^{45}\,\mathrm{erg}. \tag{29}$$

Relative to a low-energy ejection scale of

$$E_{\mathrm{ref}} = 10^{46}\,\mathrm{erg}, \tag{30}$$

this gives

$$\frac{E_{\mathrm{kin}}}{E_{\mathrm{ref}}} = 0.994. \tag{31}$$

The low ejection energy is therefore internally consistent with the observed gas mass and velocity. In the CPTG interpretation, this is important because the event need not eject a normal supernova-scale envelope. A smaller physical core plus exterior polarization can produce a strong apparent gravitational remnant without requiring the apparent mass to be baryonic material that must be energetically assembled or expelled.

6 Dust-Shell and Time-Scale Closure

The dust-shell scale reported from the remnant modeling is approximately

$$40\,\mathrm{au} \lesssim R_{\mathrm{dust}} \lesssim 200\,\mathrm{au}. \tag{32}$$

A simple travel-distance estimate for gas moving at $100\,\mathrm{km\,s^{-1}}$ over a decade is

$$R_{\mathrm{travel}} = v_{\mathrm{gas}} \Delta t. \tag{33}$$

Taking

$$\Delta t = 10.2\,\mathrm{yr}, \tag{34}$$

and converting to astronomical units gives

$$R_{\mathrm{travel}} \simeq 215\,\mathrm{au}. \tag{35}$$

This lies just above the quoted 40–200 au dust-shell interval and is close enough to be dynamically consistent at the order-of-magnitude level, given that the observed remnant structure is not a single thin spherical shell. In the CPTG reading, the shell is not merely a passive ejecta radius. It is part of the organized post-collapse environment in which curvature transport can guide anisotropic structure while polarization supplies the static apparent-mass enhancement.

7 Bolometric and X-ray Closure

The reported late bolometric luminosity is approximately

$$\log\left(\frac{L}{L_\odot}\right) \approx 3.88. \tag{36}$$

This corresponds to

$$L = 10^{3.88} L_\odot \simeq 7.59 \times 10^3 L_\odot. \tag{37}$$

Taking a progenitor luminosity scale of approximately

$$\log\left(\frac{L_{\rm prog}}{L_\odot}\right) \approx 5, \tag{38}$$

the remnant-to-progenitor luminosity fraction is

$$\frac{L}{L_{\rm prog}} \simeq 10^{-1.12} \simeq 0.076. \tag{39}$$

Thus the source sits near a seven-to-eight percent luminosity remnant level, consistent with the reported continued fading. The X-ray upper limit is

$$L_X \lesssim 1.5 \times 10^{35}\,{\rm erg\,s^{-1}}. \tag{40}$$

Using

$$L_\odot = 3.828 \times 10^{33}\,{\rm erg\,s^{-1}}, \tag{41}$$

the bolometric luminosity is

$$L_{\rm bol} \simeq 2.90 \times 10^{37}\,{\rm erg\,s^{-1}}. \tag{42}$$

The X-ray-to-bolometric upper ratio is therefore

$$\frac{L_X}{L_{\rm bol}} \lesssim 5.2 \times 10^{-3}. \tag{43}$$

This weak X-ray fraction is a key discriminator. In the conventional fallback picture, the X-ray silence must be explained by obscuration, inefficient accretion, low radiative efficiency, or transport of the emission into infrared wavelengths [2]. In the CPTG interpretation, the X-ray silence is not incidental. It is consistent with a remnant whose apparent mass is partly geometric and whose high-gradient compact core is not equivalent to a normal exposed five-solar-mass accreting black hole.

8 CPTG Phase Classification

The combined observational sequence is written as

$$\mathcal{S}_{\rm obs} = \left\{ \begin{array}{c} \text{IR brightening,} \\ \text{optical disappearance,} \\ \text{red MIR remnant,} \\ \text{molecular shell,} \\ \text{weak X rays} \end{array} \right\}. \tag{44}$$

The CPTG structural interpretation is

$$\mathcal{S}_{\rm CPTG} = \left\{ \begin{array}{c} M_{\rm core}, \chi_{\rm grad}, A_{\rm pol}, \\ M_{\rm geom}, \Sigma_{\rm trans} \end{array} \right\}. \tag{45}$$

Here $\Sigma_{\rm trans}$ denotes the effective transport/suppression state of the organized remnant environment. The object is classified as a structural-threshold candidate when the following conditions are satisfied:

$$M_{\rm core} < M_{\rm apparent}, \tag{46}$$

$$A_{\rm pol} > 1, \tag{47}$$

$$E_{\rm kin} \sim 10^{46}\,\text{erg}, \tag{48}$$

$$\frac{L}{L_{\rm prog}} \ll 1, \tag{49}$$

$$\frac{L_X}{L_{\rm bol}} \ll 1, \tag{50}$$

$$\chi_{\rm grad} \to \chi_{\rm crit}. \tag{51}$$

For M31-2014-DS1, the available data support the classification

$$\text{CPTG mass-energy-SED structural-threshold candidate.} \tag{52}$$

This phrase is intentionally limited. It does not claim a unique proof of CPTG compact-object physics. It states that the object remains coherent when interpreted as a smaller finite core with a larger apparent gravitational mass supplied by curvature polarization and organized remnant transport.

9 Comparison with Standard Interpretations

The standard failed-supernova interpretation is physically meaningful and observationally motivated. The event faded dramatically, did not show a normal supernova, and can be modeled as weak mass ejection plus fallback onto a newly formed black hole [1, 2]. The main stress is that the persistent infrared source and lack of X-rays require additional assumptions about obscuration, accretion efficiency, dust geometry, and remnant evolution.

The dusty-survivor or merger interpretation is also observationally meaningful. The persistent infrared source, asymmetric dust, and lack of X-rays can be read as evidence that a luminous central source may survive behind dust rather than as a newly formed accreting black hole [3]. The main stress is that the source must remain optically hidden while continuing to show the observed fading and red infrared behavior.

CPTG does not require the conventional alternatives to be treated as equal theoretical foundations. They are comparison interpretations. CPTG supplies a distinct geometric reading: the apparent mass, physical core mass, remnant luminosity, X-ray silence, and dust geometry need not be forced into a single ordinary-matter remnant model. The finite core and exterior gravitational appearance are structurally separated.

Observable	Standard failed SN	Dusty survivor/merger	CPTG structural collapse
Optical disappearance	Terminal collapse	Dust obscuration	Structural threshold plus obscuration
Persistent MIR source	Fallback-powered remnant	Surviving dusty source	Transported dusty remnant environment
No X-rays	Obscured or inefficient accretion	No exposed accreting BH	Suppressed finite-core/transport environment
Apparent $5M_\odot$ mass	Physical BH mass	Not required as BH mass	Apparent gravitational mass
Physical compact core	$5M_\odot$	Uncertain or none	1.1–$1.4M_\odot$ candidate band
Extra apparent mass	None	None	Curvature-polarization equivalent

Table 3: Interpretive comparison for the M31-2014-DS1 remnant.

10 Falsifiability

The CPTG interpretation makes clear observational expectations. It is strengthened if continued monitoring shows that the source remains optically absent, continues to fade or plateau in the infrared as a red dusty remnant, and remains weak in X-rays. It is stressed if the central source reappears as an ordinary luminous stellar survivor with a normal photospheric SED and no need for terminal collapse or structural horizon closure.

The immediate falsification tests are

$$\text{optical/NIR recovery} \rightarrow \text{stress for terminal CPTG collapse,} \tag{53}$$

$$\text{normal stellar SED reappearance} \rightarrow \text{stress for finite-core remnant,} \tag{54}$$

$$\text{bright persistent X rays} \rightarrow \text{stress for transport-suppressed remnant,} \tag{55}$$

$$\text{continued red MIR fading} \rightarrow \text{support for structural remnant,} \tag{56}$$

$$\text{no optical survivor recovery} \rightarrow \text{support for terminal disappearance.} \tag{57}$$

The mass split introduced here should not be read as a claim that a dynamically measured remnant mass has been revised. Rather, the conventional remnant mass is treated as a comparison-layer apparent mass inferred within ordinary collapse modeling. CPTG then asks whether the same observational record can be represented by a smaller physical finite core plus exterior curvature-polarization support and organized transport structure.

A decisive CPTG development would require a fully predictive compact-object exterior metric, a derived relation between $A_{\rm pol}$ and $\chi_{\rm grad}$, and a transport model for the remnant envelope. This article therefore treats the present calculation as a mass-energy-SED closure test, not as a completed compact-object solution.

11 Conclusion

M31-2014-DS1 is a strong single-object target for CPTG compact-object physics. The observed sequence of mid-infrared brightening, optical disappearance, persistent red infrared remnant, molecular and dusty shell structure, weak ejection energy, and X-ray silence is naturally aligned with the CPTG class of structural-threshold collapse candidates.

The most important result is the mass split. The conventional remnant mass near $5M_\odot$ need not be the physical compact-core mass. In CPTG, it can be treated as an apparent gravitational mass:

$$M_{\rm apparent} \simeq 5M_\odot. \tag{58}$$

A representative CPTG finite core in the range

$$1.1M_\odot \le M_{\rm core} \le 1.4M_\odot \tag{59}$$

requires

$$3.571 \leq A_{\mathrm{pol}} \leq 4.545, \tag{60}$$

with approximately 72–78 percent of the apparent gravitational mass supplied by exterior geometric response rather than physical baryonic core mass. The central reference case,

$$M_{\mathrm{core}} = 1.25 M_{\odot}, \tag{61}$$

gives

$$A_{\mathrm{pol}} = 4.00, \tag{62}$$

and

$$M_{\mathrm{geom}} = 3.75 M_{\odot}. \tag{63}$$

The observed gas kinetic energy scale remains close to 10^{46} erg, the dust-shell scale is dynamically compatible with decade-scale expansion at roughly $100\,\mathrm{km\,s^{-1}}$, and the X-ray-to-bolometric upper ratio remains below approximately 5.2×10^{-3}. These results keep the smaller-core CPTG interpretation physically coherent.

The object should now be treated as a benchmark CPTG vanishing-star candidate. The next decisive observations are continued JWST mid-infrared monitoring, deep optical/NIR survivor searches, and future X-ray monitoring as the dust and gas column evolve.

References

[1] K. De, M. MacLeod, J. E. Jencson, E. Lovegrove, A. Antoni, E. Kara, M. M. Kasliwal, R. M. Lau, A. Loeb, M. Masterson, A. M. Meisner, C. Panagiotou, E. Quataert, and R. Simcoe, "Disappearance of a massive star in the Andromeda Galaxy due to formation of a black hole," *Science*, 391, 689–693, 2026; arXiv:2410.14778; doi:10.1126/science.adt4853.

[2] K. De, M. MacLeod, J. E. Jencson, R. M. Lau, A. Antoni, M. J. Colmenares, J. Huang, M. Masterson, V. R. Karambelkar, M. M. Kasliwal, A. Loeb, C. Panagiotou, and E. Quataert, "Fading into darkness: A weak mass ejection and low-efficiency fallback accompanying black hole formation in M31-2014-DS1," 2026; arXiv:2601.05774; doi:10.48550/arXiv.2601.05774.

[3] E. R. Beasor, N. Smith, J. Pearson, B. Subrayan, E. Berger, D. J. Sand, and J. Strader, "The fate of the failed supernova candidate M31-2014-DS1," *Monthly*

Notices of the Royal Astronomical Society, accepted, 2026; arXiv:2601.05317; doi:10.48550/arXiv.2601.05317.

[4] Carter L Glass Jr, *Curvature-Bounded Compact Objects and the Lower Mass Gap in CPTG*, 2026.

[5] Carter L Glass Jr, *Curvature Polarization Transport Gravity: A Variational Framework for Galaxies and Cluster Mergers*, DOI: 10.13140/RG.2.2.30385.95841/1.

[6] CPTG, Supporting Python Models, Benchmark Implementations, and Research References for Curvature Polarization Transport Gravity, companion resource for the present work, available at `https://github.com/CLG2025/CPTG`.